Decimal Place Value

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS decimal *(tau ā-ira)* · decimal point *(ira)* · tenths *(hautekau)* · hundredths *(haurau)*

A Tenths, hundredths and thousandths ●○○

Remember: The first place after the decimal point is tenths, the second is hundredths and the third is thousandths. In 3.47 the 4 is worth 4 tenths, or 0.4.

1. Write the value of the underlined digit.

a) $3.\underline{4}7$ _____ **b)** $0.8\underline{5}$ _____ **c)** $12.\underline{6}$ _____

d) $4.27\underline{5}$ _____ **e)** $0.30\underline{9}$ _____ **f)** $2\underline{5}.18$ _____

g) $6.0\underline{4}$ _____ **h)** $0.\underline{5}12$ _____ **i)** $7.9\underline{3}6$ _____

B Decimals as fractions ●●○

Remember: One decimal place means tenths, two mean hundredths and three mean thousandths: $0.7 = \frac{7}{10}$, $0.23 = \frac{23}{100}$ and $0.009 = \frac{9}{1{,}000}$.

2. Write each decimal as a fraction.

a) 0.7 = _____ **b)** 0.3 = _____ **c)** 0.23 = _____ **d)** 0.09 = _____

e) 0.61 = _____ **f)** 0.009 = _____ **g)** 0.127 = _____ **h)** 0.045 = _____

3. Write each fraction as a decimal.

a) $\frac{3}{10}$ = _____ **b)** $\frac{9}{10}$ = _____ **c)** $\frac{47}{100}$ = _____ **d)** $\frac{7}{100}$ = _____

e) $\frac{81}{100}$ = _____ **f)** $\frac{3}{1{,}000}$ = _____ **g)** $\frac{259}{1{,}000}$ = _____ **h)** $\frac{31}{1{,}000}$ = _____

C Comparing and ordering decimals ●●○

⚠ **Watch out:** A longer decimal is not always larger: 0.5 is greater than 0.45, because 0.5 is the same as 0.50.

4. Write <, > or = in each box.

a) $0.45 \square 0.5$ **b)** $3.2 \square 3.18$ **c)** $0.7 \square 0.70$ **d)** $1.09 \square 1.1$

e) $0.305 \square 0.35$ **f)** $12.4 \square 12.39$ **g)** $0.099 \square 0.1$ **h)** $6.25 \square 6.250$

5. Write each list in order, from smallest to largest.

a) 0.5, 0.45, 0.54, 0.405 _____ **b)** 3.2, 3.02, 3.22, 2.3 _____

c) 0.09, 0.9, 0.1, 0.19 _____ **d)** 7.5, 7.05, 7.55, 7.505 _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Multiplying and dividing by 10, 100 and 1,000 ●●●

⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

6. Work out.

a) 45×100 = _____ **b)** 3.7×10 = _____ **c)** $0.62 \times 1{,}000$ = _____ **d)** 208×10 = _____

e) $4{,}800 \div 100$ = _____ **f)** $56 \div 10$ = _____ **g)** $7 \div 100$ = _____ **h)** $930 \div 1{,}000$ = _____

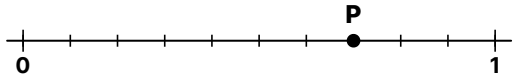
i) 1.05×100 = _____ **j)** 0.4×10 = _____ **k)** $2.5 \div 10$ = _____ **l)** $69 \div 100$ = _____

E Decimals on a number line ●●●

💡 **Remember:** First work out what each small step on the line is worth.

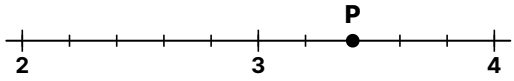
7. Write the number that the point P shows.

a)



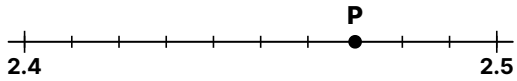
$P =$ _____

b)



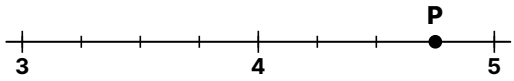
$P =$ _____

c)



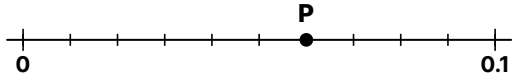
$P =$ _____

d)



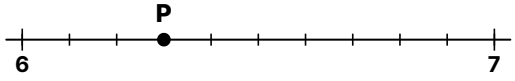
$P =$ _____

e)



$P =$ _____

f)



$P =$ _____

★ Extension

8. Write a number that is between each pair of numbers.

a) 0.3 and 0.4 **b)** 2.5 and 2.6 **c)** 0.07 and 0.08

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Decimal Place Value

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS decimal *(tau ā-ira)* · decimal point *(ira)* · tenths *(hautekau)* · hundredths *(haurau)*

A Tenths, hundredths and thousandths ●○○

Remember: The first place after the decimal point is tenths, the second is hundredths and the third is thousandths. In 3.47 the 4 is worth 4 tenths, or 0.4.

1. Write the value of the underlined digit.

a) $5.\underline{2}9$ _____ **b)** $0.6\underline{3}$ _____ **c)** $14.\underline{8}$ _____

d) $2.46\underline{1}$ _____ **e)** $0.70\underline{5}$ _____ **f)** $3\underline{8}.24$ _____

g) $9.0\underline{7}$ _____ **h)** $0.\underline{3}84$ _____ **i)** $1.6\underline{5}2$ _____

B Decimals as fractions ●●○

Remember: One decimal place means tenths, two mean hundredths and three mean thousandths: $0.7 = \frac{7}{10}$, $0.23 = \frac{23}{100}$ and $0.009 = \frac{9}{1{,}000}$.

2. Write each decimal as a fraction.

a) 0.9 = _____ **b)** 0.1 = _____ **c)** 0.37 = _____ **d)** 0.03 = _____

e) 0.59 = _____ **f)** 0.007 = _____ **g)** 0.213 = _____ **h)** 0.061 = _____

3. Write each fraction as a decimal.

a) $\frac{7}{10}$ = _____ **b)** $\frac{5}{10}$ = _____ **c)** $\frac{29}{100}$ = _____ **d)** $\frac{1}{100}$ = _____

e) $\frac{73}{100}$ = _____ **f)** $\frac{9}{1{,}000}$ = _____ **g)** $\frac{467}{1{,}000}$ = _____ **h)** $\frac{53}{1{,}000}$ = _____

C Comparing and ordering decimals ●●○

⚠ **Watch out:** A longer decimal is not always larger: 0.5 is greater than 0.45, because 0.5 is the same as 0.50.

4. Write <, > or = in each box.

a) $0.6 \square 0.58$ **b)** $4.17 \square 4.2$ **c)** $0.90 \square 0.9$ **d)** $2.1 \square 2.08$

e) $0.46 \square 0.406$ **f)** $15.29 \square 15.3$ **g)** $0.2 \square 0.199$ **h)** $3.40 \square 3.4$

5. Write each list in order, from smallest to largest.

a) 0.6, 0.56, 0.65, 0.506 _____ **b)** 4.1, 4.01, 4.11, 1.4 _____

c) 0.08, 0.8, 0.2, 0.18 _____ **d)** 2.5, 2.05, 2.55, 2.505 _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Multiplying and dividing by 10, 100 and 1,000 ●●●

⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

6. Work out.

a) 63×100 = _____ **b)** 5.2×10 = _____ **c)** $0.48 \times 1{,}000$ = _____ **d)** 307×10 = _____

e) $6{,}500 \div 100$ = _____ **f)** $84 \div 10$ = _____ **g)** $9 \div 100$ = _____ **h)** $710 \div 1{,}000$ = _____

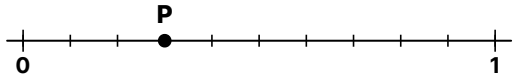
i) 2.06×100 = _____ **j)** 0.7×10 = _____ **k)** $3.5 \div 10$ = _____ **l)** $48 \div 100$ = _____

E Decimals on a number line ●●●

💡 **Remember:** First work out what each small step on the line is worth.

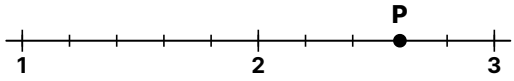
7. Write the number that the point P shows.

a)



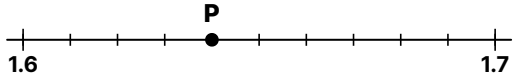
$P =$ _____

b)



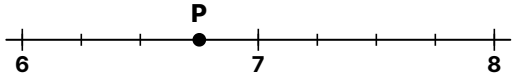
$P =$ _____

c)



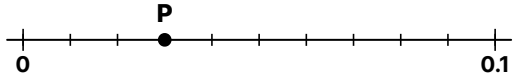
$P =$ _____

d)



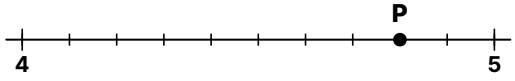
$P =$ _____

e)



$P =$ _____

f)



$P =$ _____

★ Extension

8. Write a number that is between each pair of numbers.

a) 0.6 and 0.7 **b)** 3.1 and 3.2 **c)** 0.04 and 0.05

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.